

Angel Rodriguez

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Science

Organizations: KnightHacks, SHPE, ColorStack

Orlando, FL

Expected Graduation: Spring 2028

EXPERIENCE

Software Engineering Intern

Jan. 2026 – Apr. 2026

Minoria Tech

Orlando, FL

- Designed and implemented **authentication systems and database architecture** supporting CRM workflows for enterprise clients including Lockheed Martin, Dell, and Red Hat.
- Developed **RESTful APIs** using **FastAPI and SQL**, replacing third-party CRM dependencies and improving system control and scalability.
- Collaborated in a Git-based team environment, contributing to code reviews and API design documentation to ensure maintainability and efficient onboarding

NASA L'SPACE NPWEE Member

Jan. 2026 – Mar. 2026

L'SPACE

Remote

- Selected for NASA's competitive L'SPACE program, contributing to a mission proposal focused on low-temperature Sabatier reactors for Mars ISRU fuel production.
- Collaborated on a Nickel-Manganese catalyst system converting CO₂ and H₂ into methane, targeting an operating-temperature reduction from **~400°C to ~250°C**.

Robotics Team Lead Programmer

Sep. 2023 – June. 2025

Botball

New Jersey

- Led programming team of **10** for an autonomous robot using C and KIPR Wombat Controller, enabling full task automation in competition scenarios
- Integrated sensors, camera vision, servos, and motor control systems to achieve **75% task success rate** in autonomous navigation and object manipulation challenges
- Developed and tested modular C code for obstacle navigation and object handling, improving execution consistency and reliability across competition runs

PROJECTS

DSL – AI Humanitarian Crisis Intelligence Platform | *Python, FastAPI, React, Databricks, OpenAI, Pandas*

- Won **3rd of 24** in Georgia Tech's Databricks Track for an AI platform identifying underfunded humanitarian crises.
- Led backend development, building **FastAPI/Express and Pandas** pipelines powering **6+ endpoints** for crisis rankings, funding gaps, structural mismatch, and funding velocity.
- Built an **OpenAI RAG analyst** grounding natural-language queries in crisis data and **5+ derived metrics**; automated alerts for severity ≥ 4.0 and $< 25\%$ funding coverage.
- Integrated FastAPI services with a **React frontend**, transforming backend crisis data into interactive Mapbox visualizations, country drill-downs, funding comparisons, and real-time analytical dashboards.

Halo — Smart Cycling Safety Helmet — ShellHacks 2026 | *Python, Raspberry Pi, YOLO, SwiftUI, OpenCV*

- 1st out of 77 teams** — Waymo Mobility Challenge and **1st out of 34 teams** — State Farm Challenge at ShellHacks 2026, Florida's largest hackathon.
- Led **iOS and systems integration** for a SwiftUI companion app, connecting MapKit navigation, GPS-aware search, Raspberry Pi SSE telemetry, CV event state, incident playback, and haptic alerts into a real-time safety interface operating at **~30 FPS telemetry updates**.
- Collaborated on a **dual-camera Raspberry Pi 5 perception pipeline** using YOLO26n $\rightarrow$ NCNN, object tracking, TTC-based hazard classification, ultrasonic sensor fusion, and Arduino-driven alerts, achieving **~18–21 ms detection latency** during live testing.
- Synchronized **4 critical alert channels**—helmet display, ElevenLabs audio, mobile UI, and iPhone haptics—for front-obstacle BRAKE events; integrated incident capture and post-event analysis.

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL, HTML/CSS

Frameworks and Data: React, FastAPI, Node.js, SQLite, Pandas, Tailwind CSS

AI: PyTorch, OpenCV, YOLO, RAG, Gemini, Claude Code, Codex

Developer Tools: Linux, Git, Vercel, VS Code